* Syntax-:

int[] a = new int[5]

* Count greater than itself-:

	0	1	2	3	4	5	6
a= [	3,	-2,	6,	8,	4,	8,	5]

ans => 5

* observation-:

→ for every max element there won't be any element greater than it

→ for every other element, there will be atmost one element greater than it

1) Iterate & get max value
2) Iterate & get count of elements != max

TC: O(N)

SC: O(1)

* Check Pairs (i, j) -:

such that

$a[i] + a[j] = k$ & $i \ne j$

$a = [3 \quad -2 \quad 1 \quad 4 \quad 5 \quad 6 \quad 8]$

$k = 10$

$4 + 6 = 10$ true

Idea -:

→ Run loop with i for all indices

→ Run a nested loop with j over all indices.

TC: $O(N^2)$

SC: $O(1)$

* Reverse entire Array -:

arr[] =	[-1	4	7	6	-2	7	8	10]
	0	1	2	3	4	5	6	7
	i							j

$i < j$

TC: $O(\log n)$

SC: $O(1)$

* Reverse part of Array (S to E)

a[] = [-3 4 2 8 7 9 6 2 10]

0 1 2 3 4 5 6 7 8

[S E] = [3 7]

TC: O(N)

SC: O(1)

* Rotate Array from last to first by K times-: (Note-: K > 0)

K = 5

$a[13] = [a_0\ a_1\ a_2\ a_3\ a_4\ a_5\ a_6\ a_7\ a_8\ a_9\ a_{10}\ a_{11}\ a_{12}]$

Reverse the entire array

$[a_{12}\ a_{11}\ a_{10}\ a_9\ a_8\ a_7\ a_6\ a_5\ a_4\ a_3\ a_2\ a_1\ a_0]$

reverse the first K elements

Reverse the last (N-K) elements

rotate by 5 times-:

$a_8\ a_9\ a_{10}\ a_{11}\ a_{12}\ a_0\ a_1\ a_2\ a_3\ a_4\ a_5\ a_6\ a_7$

1) Reverse entire array, s = 0, e = N-1

2) Reverse first K elements, s = 0, e = K-1

3) Reverse last N-K elements, s=k

e=N-1

$$\boxed{\begin{array}{l} \text{if} \quad k > N \\ \quad\quad k = k \% N \end{array}}$$

TC: O(N)

SC: O(1)